

Administrator Guide: Academic Advisement System

Illustrated curriculum-management and operations manual Version 1.0 | Academic Advisement System prototype

This guide covers role-based access, database inspection, Major Constructor drafts, curriculum bins, alternatives, prerequisites, elective pools, Core adjustments, preview, validation, review, publication, equivalencies, AI settings, and quality assurance.

Change-control rule: Treat every published curriculum as institutional data. Use official sources, keep drafts reviewable, validate before approval, and preserve a rollback path.

AA

Academic Advisement

Plan with a complete curriculum view

ProgramsMy ProgressCompareAdminLog out

Admin Dashboard

Administrative dashboard for curriculum and advising management.

Database Summary

26

Institutions

51

Departments

142

Majors

1410

Courses

9

Equivalency Rules

Curriculum Studio

Major Constructor

Create draft programs and concentrations with credit-counted curriculum bins.

Open Constructor

Database

Institutions

View, add, edit, and safely delete colleges or institutions such as BMCC.

Manage Table

Database

Departments

View academic departments connected to programs and curriculum structures.

View Table

Programs

Majors / Programs

View academic programs such as CS, CIS, CNT, catalog years, and degree types.

View Table

Curriculum

Courses

View course codes, titles, credits, and database course records.

Requirements

Requirement Groups

View groups such as Program Requirements, Program Electives, Common Core, and

Transfer Rules

Course Equivalencies

Approve, edit, and deactivate directional course-transfer mappings between institutions.

Logic

Prerequisites

Future editor for prerequisite relationships, alternatives, and course eligibility rules.

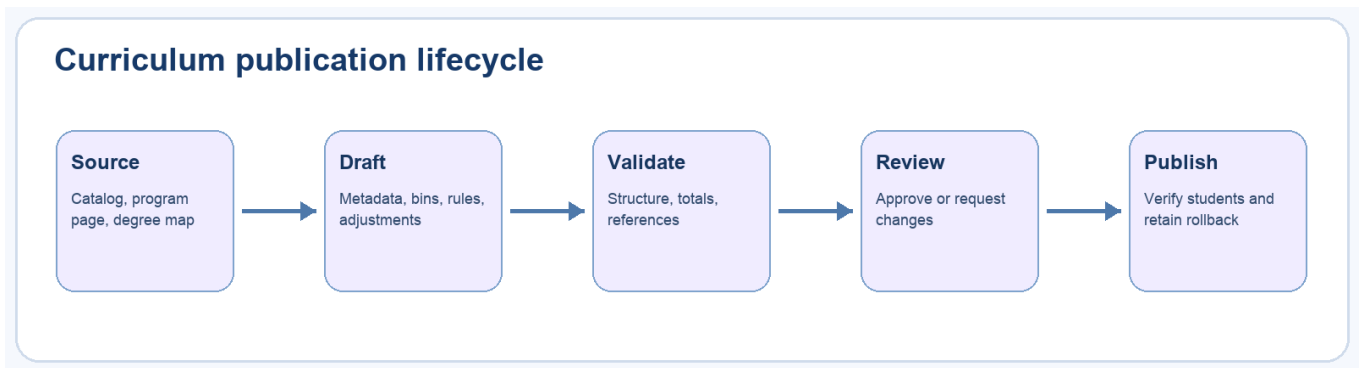
Administrator dashboard

1. Administrator responsibilities

Administrators maintain the application representation of academic policy. Typical responsibilities include:

- Managing institutions, departments, programs, courses, and requirement groups.
- Creating and reviewing program drafts.
- Translating official requirements and footnotes into machine-enforced rules.
- Maintaining course equivalencies with provenance and status.
- Verifying student-facing selectors, modals, progress, and plans.
- Configuring the AI advisor without weakening source restrictions.
- Reviewing logs and identifying data-quality or usability problems.

An administrator should not invent missing curriculum rules. Mark uncertain values for review and cite the official catalog, department page, degree map, or articulation source.



Administrator workflow

2. Sign in and verify administrator access

- 1 Open the login page.
- 2 Enter the administrator credentials.
- 3 Select **Log In**.
- 4 Confirm that **Admin** appears in the main navigation.
- 5 Open **Admin** and verify the dashboard loads.

The default local account is admin / admin. Override APP_USERNAME, APP_PASSWORD, and SESSION_SECRET in hosted environments.

Administrative access is enforced server-side. Hiding the navigation alone is not considered authorization.

Use a tester account for student-experience verification so an administrator-only link or API dependency does not accidentally enter the student workflow.

3. Read the Admin Dashboard

The dashboard groups tools by operational purpose:

- **Major Constructor** for draft-based curriculum authoring.
- **Institutions** and **Departments** for organizational records.
- **Majors / Programs** and **Courses** for catalog data.
- **Requirement Groups** for curriculum structure.
- **Course Equivalencies** for transfer rules.
- **Advisor Chat Logs** for AI interaction review.
- Future cards identify planned but unavailable interfaces.

Check the database summary before large edits. Unexpectedly low counts can indicate a failed seed or incorrect database connection.

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Admin Dashboard

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4. Inspect database tables safely

Dashboard table views help administrators audit the live records without editing CSV files blindly.

When inspecting a course, verify:

- Institution
- Course code and normalized display code
- Title
- Credits
- Department relationships
- Placeholder or choice-group code when applicable

For programs and requirements, verify catalog year, degree type, group category, credit target, ordering, and official source.

Avoid deleting records that may be referenced by programs, drafts, prerequisites, alternatives, or equivalencies. Prefer deactivation or a new catalog version when historical data must remain reproducible.

AA

Academic Advisement

Plan with a complete curriculum view

ProgramsMy ProgressCompareAdminLog out

Id	Code	Title	Credits	Choice Group Code	Institution Codes	Department Ids
5	ART 100	Introduction to Art	3	—	BMCC	
16	BIO 105	Human Biology	3	—	BMCC	
393	CCNY-TECH-ELECTIVE-1	Approved Technical Elective I	3	—		
394	CCNY-TECH-ELECTIVE-2	Approved Technical Elective II	3	—		
521	CS-GENERAL	General Elective or Common Core STEM excess credits	4	BMCC_GENERAL_ELECTIVE		
43	FC-SCI-WORLD	Scientific World Requirement	6	FC_SCIENTIFIC_WORLD		
7	MUS 100	Introduction to Music	3	—	BMCC	
18	ACC 122	Accounting Principles I	3	—	BMCC	1, 5, 6
1107	ACC 150	Personal Accounting and Financial Literacy	3	—	BMCC	1
1095	ACC 222	Accounting Principles II	3	—	BMCC	1, 5
1096	ACC 231	Accounting Applications on Microcomputers	3	—	BMCC	1
1097	ACC 241	Taxation: Federal	3	—	BMCC	1
1108	ACC 242	Taxation of Business Entities	3	—	BMCC	1
1098	ACC 330	Intermediate Accounting I	3	—	BMCC	1
1109	ACC 331	Accounting Information Systems	3	—	BMCC	1

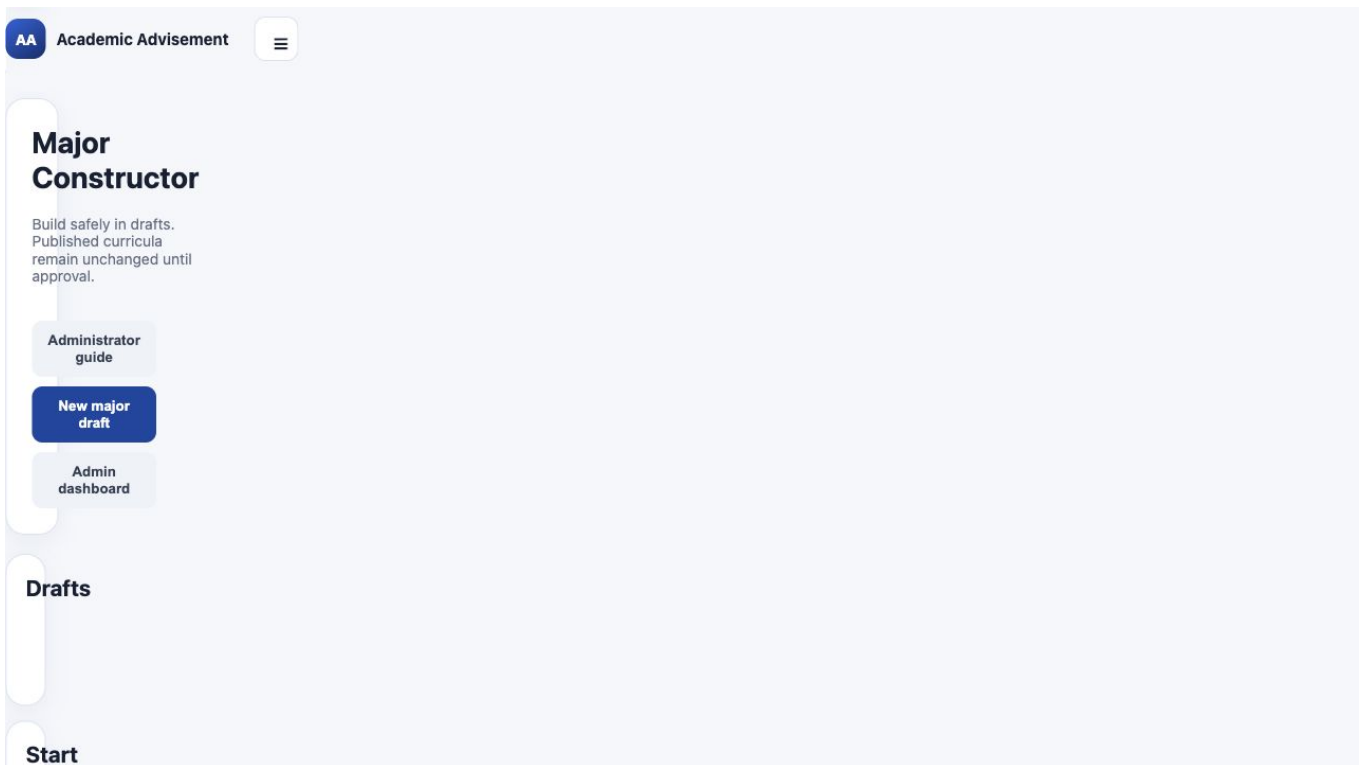
Course database table

5. Start a Major Constructor draft

- 1 Open **Major Constructor**.
- 2 Review the existing draft list.
- 3 Select **New major draft**.
- 4 Confirm the new item appears with draft status.
- 5 Enter metadata before adding large numbers of courses.

Drafts isolate work from published curricula. Creating or saving a draft should not alter the student selector.

Use a descriptive draft name early. Avoid multiple indistinguishable **Untitled program** records because reviewers cannot tell which draft is current.



Empty Major Constructor

6. Enter program metadata

Complete these fields:

- **Campus** - controls course availability and default Core groups.
- **Department** - organizes the program and prioritizes course search.
- **Program name** - student-facing official name.
- **Program code** - stable internal code.
- **Degree type** - for example A.A., A.S., A.A.S., B.A., or B.S.
- **Catalog year** - the authoritative curriculum period.
- **Official source URL** - department, catalog, or program requirements page.

After selecting a campus, verify that only that campus's courses are available. If the campus changes later, re-audit every course and rule.

The screenshot shows the 'Major Constructor' interface. At the top, there's a navigation bar with 'AA Academic Advisement' and a subtitle 'Plan with a complete curriculum view'. To the right are links for 'Programs', 'My Progress', 'Compare', 'Admin', and 'Log out'. Below this, the 'Major Constructor' section has a subtitle 'Build safely in drafts. Published curricula remain unchanged until approval.' and buttons for 'Administrator guide', 'New major draft', and 'Admin dashboard'. On the left, a 'Drafts' sidebar shows 'Untitled program' with 'draft - 0 versions'. The main form area is titled 'Untitled program' and includes a 'DRAFT' label and buttons for 'Save draft', 'Create version', and 'Delete draft'. The form fields are: 'Campus' (dropdown), 'Department' (dropdown), 'Program name' (text input with 'Untitled program'), 'Program code' (text input with 'PSY-AS'), 'Degree type' (text input with 'A.S.'), 'Catalog year' (text input with '2026-2027'), 'Official source URL' (text input with 'https://...'), and 'Concentrations' (a 'General' button).

Major Constructor metadata form

7. Represent concentrations

Use concentration tabs when one program has distinct named curricula.

- 1 Keep **General** only when it is an actual general pathway.
- 2 Enter the concentration name and select **Add concentration**.
- 3 Use **Rename active** for corrections.
- 4 Use **Remove active** only after verifying that no required content will be lost.

For student-facing clarity, separate concentrations may also be published as distinct program choices when they have materially different requirements, such as Psychology General and Psychology STEM.

Do not create a concentration merely for an informal recommendation. The name and requirements should be supported by an official source.

8. Understand the four curriculum bins

The constructor uses four destination bins:

- **Major requirements** - specifically required curriculum courses.
- **Major electives** - controlled pools or elective placeholders.
- **Common Core** - required Pathways categories or institution-specific equivalents.
- **Flexible Core** - approved Flexible Core areas.

Each bin tracks available credits plus required credit and course counts. The count at the top is a validation aid, not a substitute for checking official totals.

Campus selection can prepopulate Common and Flexible Core groups. Administrators then add program-specific adjustments rather than rebuilding canonical pools for every major.

9. Add courses in List view

- 1 Select the destination bin.
- 2 Search by course code or title.
- 3 Confirm the course belongs to the selected campus.
- 4 Select the course card to add it.
- 5 Review the bin's credit total.
- 6 Remove incorrect cards with the remove control.

The search ranks the selected department's courses first but still permits relevant courses from other departments at the same campus.

Add placeholders only when they connect to a non-empty choice group. A selectable placeholder with no group code produces an incomplete student selector.

AA **Academic Advisement**
Plan with a complete curriculum view

ProgramsMy ProgressCompareAdminLog out

Major Constructor

Build safely in drafts. Published curricula remain unchanged until approval.

Administrator guideNew major draftAdmin dashboard

Drafts
Untitled program
draft - 0 versions

DRAFT

Save draftCreate versionDelete draft

Untitled program

Campus
Borough of Manhattan Community College

Department
Computer Information Systems

Program name
Sample Computing Program

Program code
SAMPLE-AS

Degree type
A.S.

Catalog year
2026-2027

Official source URL
https://...

Concentrations

General

e.g. STEM ConcentrationAdd concentrationRename activeRemove active

Curriculum bins

Select a destination, then click courses to add them. Credit totals update immediately.

ListBoard

Major Constructor list view

10. Use Board view

Select **Board** to display all four bins in a compact visual workspace. Drag course cards from the course library into a destination bin or between bins.

Use Board view for curriculum organization and comparison. Use List view when entering detailed credit targets or reviewing long course titles.

During dragging:

- The target bin changes color.
- The dragged card reflects the prospective destination.
- All four pastel-colored bins remain visible on a standard desktop width.

After a drag, verify the destination and the updated credit count. A visual move does not replace rule creation when the course is part of an alternative or elective pool.

The screenshot shows the 'Academic Advisement' interface. At the top, there's a navigation bar with 'AA Academic Advisement' and a subtitle 'Plan with a complete curriculum view'. To the right are links for 'Programs', 'My Progress', 'Compare', 'Admin', and 'Log out'. The main content area is divided into sections. The 'Official source URL' section has a text input field with 'https://...'. Below it is the 'Concentrations' section with a 'General' tab and a text input field containing 'e.g. STEM Concentration'. There are buttons for 'Add concentration', 'Rename active', and 'Remove active'. The 'Curriculum bins' section has a 'List' button and a selected 'Board' button. A descriptive text says: 'Arrange the curriculum visually across four columns. Credit totals update immediately.' Below this is a light blue instruction box: 'Drag a course from the library into a column, or move a card between columns. On a keyboard or touch screen, choose a destination and activate the course card.' A search bar contains 'CSC 210' and a dropdown menu shows 'Major requirements'. Below the search bar is a course card for 'CSC 210 3 CR Computer Programming II'. At the bottom, there are four colored bins: 'Major requirements' (purple, 7 available credits), 'Major electives' (pink, 0 available credits), 'Common Core' (green, 12 available credits), and 'Flexible Core' (yellow, 15 available credits). Each bin has a table with 'CREDITS REQUIRED' and 'COURSES REQUIRED' columns.

Major requirements		Major electives		Common Core		Flexible Core	
7 available credits		0 available credits		12 available credits		15 available credits	
CREDITS REQUIRED	COURSES REQUIRED	CREDITS REQUIRED	COURSES REQUIRED	CREDITS REQUIRED	COURSES REQUIRED	CREDITS REQUIRED	COURSES REQUIRED

Major Constructor board view

11. Create OR alternatives

Use an alternative rule when the official curriculum says one course **OR** another course satisfies one requirement.

- 1 Add both courses to an appropriate curriculum bin.
- 2 Choose **Alternative A**.
- 3 Choose **OR Alternative B**.
- 4 Select **Add OR rule**.
- 5 Preview the student interface.

Expected student behavior:

- Both options appear in one clearly grouped card or dialog.
- Selecting one marks the other as not needed or unavailable.
- The requirement receives credit only once.

Do not use an OR rule for a multi-course elective pool or a recommended course.

12. Create prerequisites and sequences

Use a prerequisite rule when one selected course must be completed before another becomes available.

- 1 Add both courses to the curriculum.
- 2 Choose the dependent course under **Course**.
- 3 Choose the prerequisite under **Requires**.
- 4 Select **Add prerequisite**.
- 5 Preview both locked and unlocked states.

For alternative prerequisites or sequences, reproduce the official logic precisely. Do not infer prerequisites solely from a degree-map order; confirm them in an authoritative catalog or course source.

A course selected as completed should synchronize across every requirement location, and dependent courses should update immediately.

13. Create an elective pool

Use an elective pool when students choose from multiple approved courses.

Define:

- A clear student-facing pool name.
- The destination bin.
- Credits required.
- Courses required when the rule specifies a count.
- Every approved course in the pool.
- Additional level, subject, or distribution rules when supported.

Only courses participating in the curriculum should be available to rule selectors. This prevents a hidden course from affecting a visible requirement.

Test the pool below, at, and above the selection limit. Confirm that a student can replace an existing choice but cannot overfill the requirement.

14. Configure Common and Flexible Core adjustments

The target architecture uses:

- 1 One canonical base-membership group for each Core area.
- 2 A program adjustment only when the major differs from the base pool.
- 3 Program CSV rows that reference base or derived choice-group codes.

In the adjustment editor:

- Select the base Core group.
- Select the placeholder requirement.
- Override credits or course count only when required.
- Limit allowed subject prefixes when the official rule does so.
- Include or exclude specific selected courses.
- Write a concise student-facing adjustment note.

The note should translate the rule, not reproduce an unreadable footnote.

15. Preview and validate

Select **Preview major** to inspect the same bin structure students will receive. Then select **Validate**.

Validation should catch:

- Missing campus, department, code, degree type, or catalog year.
- Empty required bins.
- Credit totals that do not match the declared requirement.
- Placeholder requirements with missing or empty groups.
- Rules referencing courses outside the curriculum.
- Invalid alternatives or prerequisites.
- Duplicate or conflicting program identity.

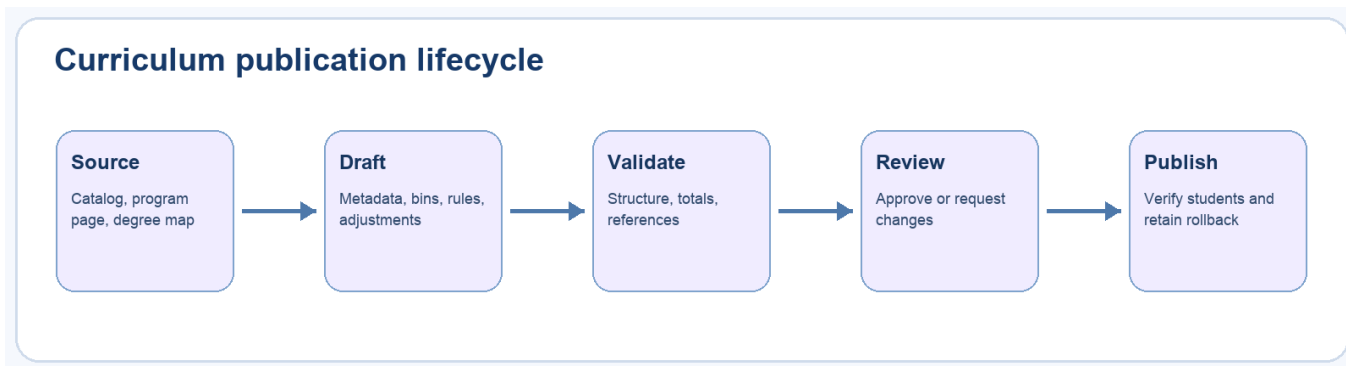
Validation success means the draft is structurally consistent; it does not prove that it matches policy. Complete an official-source audit before review.

16. Review, approve, publish, and roll back

The intended lifecycle is:

- 1 Save the working draft.
- 2 Create a version at a meaningful checkpoint.
- 3 Validate.
- 4 Submit for review.
- 5 Approve or request changes.
- 6 Publish only an approved version.
- 7 Verify the student selector and progress page.
- 8 Roll back if a published defect is found.

Separate authoring from approval whenever possible. Record the official source and review notes so another administrator can reconstruct the decision.



Draft lifecycle

17. Maintain course equivalencies

Course equivalencies are directional and institution-specific. A complete record can include:

- Source and destination institutions.
- One source course for a direct equivalency.
- Multiple source courses for a combination equivalency.
- Destination course.
- Minimum grade.
- Catalog-year bounds.
- Status such as draft, approved, inactive, or superseded.
- Official articulation source and notes.

Never infer equivalence from matching course numbers or similar titles. Keep unverified mappings out of approved student calculations.

AA

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Course Equivalencies

Add EquivalencyClose

	Id	Source Institution Id	Source Institution	Source Institution Code	Source Course Code	Source Course Codes	Source Course Title	Source Credits	Target Institution Id	Target Institution	Target Institution Code	Target Course Code	Target Course Title	Target Credits	Equivalency Type
6	2		Borough of Manhattan Community College	BMCC	CSC 111 + CSC 211	CSC 111, CSC 211	Introduction to Programming plus Advanced Programming Techniques	7	5	The City College of New York	CCNY	CSC 10300	Introduction to Computing for Majors	3	combination
7	2		Borough of Manhattan Community College	BMCC	CSC 215	CSC 215	Fundamentals of Computer Systems	3	5	The City College of New York	CCNY	CSC 21100	Fundamentals of Computer Systems	3	direct
8	2		Borough of Manhattan Community College	BMCC	CSC 231	CSC 231	Discrete Structures and Applications to Computer Science	4	5	The City College of New York	CCNY	CSC 10400	Discrete Mathematical Structures I	4	direct
9	2		Borough of Manhattan Community College	BMCC	CSC 331	CSC 331	Data Structures	3	5	The City College of New York	CCNY	CSC 21200	Data Structures	3	direct
4	2		Borough of Manhattan Community College	BMCC	ENG 101	ENG 101	English Composition	3	5	The City College of New York	CCNY	ENGL 11000	Freshman Composition	3	direct
1	2		Borough of Manhattan Community College	BMCC	MAT	MAT	Analytic	4	5	The City College of New York	CCNY	MATH	Calculus I	4	direct

Course equivalency management

18. Configure the AI advisor

Open **AI Settings** to configure:

- Agent ID and student-facing name.
- Default page name and URL.
- Gemini model.
- API key or environment-based key.
- System prompt.
- Metrics visibility.
- Advisor logging.
- Reusable embed snippet.

The prompt should require the advisor to use only page context, identify missing information, and avoid inventing requirements, exceptions, transfer rules, or guarantees.

Store production secrets in environment variables or an appropriate secrets service. Do not commit real keys.

AI Settings

Admin DB Graph Logout

Configure the reusable AI advisor agent, prompt, logging, metrics, and embed snippet.

Agent Configuration API key missing

Agent ID advisor_progress	Agent Name AI Academic Advisor
Default Page Name Degree Progress	Default Page URL /db-progress
Model gemini-2.5-flash	Gemini API Key Leave blank to keep existing key Stored server-side in config/ai_settings.json for this prototype. For production, use encrypted storage or environment variables.

System Prompt

You are an academic advising assistant embedded inside a webpage.

Use **ONLY** the page context provided by the webpage. Do not invent official degree requirements, course policies, prerequisites, transfer rules, or exceptions.

If the context does not contain enough information, say that clearly and suggest what information is missing.

Be concise, practical, and student-friendly. Explain locked courses, completed courses, not-needed alternatives, elective completion, and recommendations when relevant.

☒ Show response time / token metrics in chat
☒ Enable advisor logs

Save AI Settings Reload

Reusable Embed Snippet

Paste this into any page that has a page-specific `contextProvider`. Replace the example context with the data that page should expose to the AI advisor.

Generate Snippet Copy

Click "Generate Snippet".

AI Settings

19. Regression testing checklist

Before publishing curriculum or platform changes:

- Run the automated test suite.
- Seed a clean local database.
- Verify no duplicate program appears in the selector.
- Confirm established majors retain their previous requirements and pools.
- Test a new student and a student with completed coursework.
- Test locked prerequisites and alternative synchronization.
- Test elective limits and requirement-satisfied states.
- Test Writing Intensive confirmation.
- Generate and inspect a degree-plan PDF.
- Test major-change and transfer comparison.
- Sign in as tester and confirm admin access is unavailable.

Use a representative mature program, such as Computer Science, as a regression baseline.

20. Data-source and audit checklist

For every new or modified program, preserve:

- Official program requirements URL.
- Official catalog or SmartCatalog URL when available.
- Degree-map PDF in project assets.
- Catalog year and degree type.
- Footnote-by-footnote interpretation.
- Program requirements and electives.
- Core adjustments.
- Alternatives and prerequisites.
- Elective pool membership and limits.
- Course equivalency sources.

If sources conflict, stop publication and request department clarification. A technically valid draft can still be academically incorrect.

21. Incident response and rollback

When a published defect is reported:

- 1 Reproduce it with the same campus, program, and catalog year.
- 2 Determine whether the problem is data, rule logic, UI, authentication, or deployment.
- 3 Record screenshots and exact expected behavior.
- 4 Protect students from the incorrect result with a rollback or deactivation.
- 5 Correct the smallest authoritative source layer.
- 6 Add or update an automated regression test.
- 7 Reseed and retest affected and baseline majors.
- 8 Publish through the normal approval workflow.

Do not overwrite historical curriculum merely to repair the newest version.

Administrator quick reference

Task	Primary tool	Required verification
Inspect data	Dashboard table	Institution, code, credits, references
Create a major	Major Constructor	Metadata and official source
Add courses	List or Board	Campus, bin, totals
Create alternatives	OR rule	Mutual exclusion in preview
Create prerequisites	Prerequisite rule	Locked and unlocked states
Create electives	Elective pool	Membership and limits
Adjust Core	Core adjustment	Base group, overrides, note
Publish	Review workflow	Validation, approval, student regression
Add transfer mapping	Equivalencies	Direction, source, status
Configure AI	AI Settings	No invented policy, secrets protected
Investigate defect	Logs and reproduction	Minimal fix plus regression test

Publish only what can be traced to an authoritative source and reproduced by another reviewer.